Youwei Xiao

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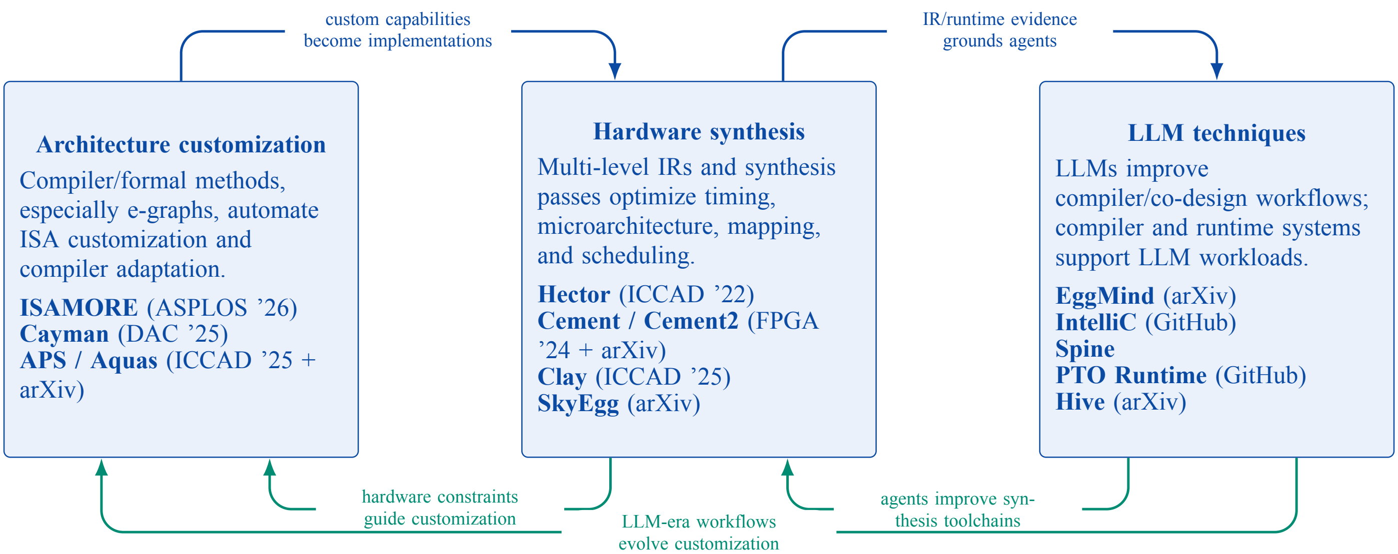
Education

Bachelor of Science degree in Computer Science — September 2018 - July 2022
School of Electronics Engineering and Computer Science, Peking University — Beijing, China

Ph.D. Candidate in EDA — September 2022 - now
School of Integrated Circuits, Peking University — Beijing, China

Research

My research builds compiler-centered hardware-software co-design systems for domain-specific computing. The central problem is how real software demands can be translated into reusable architectural capabilities, realized as high-quality hardware, and exposed back to software through compiler interfaces that remain analyzable, verifiable, and reusable.



Architecture customization. I explored compiler techniques and formal methods, especially e-graphs, to automate ISA customization and compiler adaptation. *ISAMORE* (DOI) finds reusable custom instructions by anti-unifying equivalent program fragments. *Cayman* (DOI) discovers accelerator opportunities from whole applications while optimizing control flow and data access. *APS* (DOI, GitHub) and *Aquas* (preprint) connect customized hardware capabilities to retargetable compiler support.

Hardware synthesis. I created intermediate representations at multiple abstraction levels and proposed synthesis passes that optimize timing, microarchitecture, mapping, and scheduling. *Hector* (DOI, GitHub) provides a multi-level MLIR substrate for hardware synthesis. *Cement* (DOI) and *Cement2* (preprint) make cycle timing and temporal transactions first-class FPGA design abstractions. *Clay* (DOI) synthesizes microarchitecture-aware ASIP instructions, while *SkyEgg* (preprint) uses e-graphs for joint implementation selection and scheduling.

LLM techniques. I explored both LLMs for compilers and compiler/system techniques for LLMs. *EggMind* (preprint) uses LLMs to synthesize reusable equality-saturation strategies. *IntelliC* (GitHub) studies inspectable compiler representations for human and agent collaboration. *Spine* builds agentic compiler workflows for long-horizon optimization and design adaptation. *PTO Runtime* (GitHub) executes compiled task graphs for serving on distributed Ascend architectures. *Hive* (preprint) extends the systems view to multi-agent inference infrastructure.

The closed loop is that architecture customization exposes reusable hardware capabilities, hardware synthesis turns them into efficient implementations, and LLM-era techniques advance the earlier compiler and co-design toolchains while providing evolving capabilities that make the loop practical and self-improving.

Publications

- **Youwei Xiao**, Chenyun Yin, Yitian Sun, and Yun Liang. 2026. Finding Reusable Instructions via E-Graph Anti-Unification. In Proceedings of the 31st ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS '26). doi:10.1145/3779212.3790162. **[Best Paper Award, 5/1048]**
- **Youwei Xiao**, Yuyang Zou, Yansong Xu, Yuhao Luo, Yitian Sun, Chenyun Yin, Ruifan Xu, Renze Chen, and Yun Liang. 2025. APS: Open-Source Hardware-Software Co-Design Framework for Agile Processor Specialization. In Proceedings of the 44th IEEE/ACM International Conference on Computer-Aided Design (ICCAD '25). doi:10.1109/ICCAD66269.2025.11240817. [Invited Paper]
- Weijie Peng*, **Youwei Xiao***, Yuyang Zou, Zizhang Luo, and Yun Liang. 2025. Clay: High-level ASIP Framework for Flexible Microarchitecture-Aware Instruction Customization. In Proceedings of the 44th IEEE/ACM International Conference on Computer-Aided Design (ICCAD '25). doi:10.1109/ICCAD66269.2025.11240669. (*Equal contribution)
- **Youwei Xiao**, Fan Cui, Zizhang Luo, Weijie Peng, and Yun Liang. 2025. Cayman: Custom Accelerator Generation with Control Flow and Data Access Optimization. In Proceedings of the 62nd ACM/IEEE Design Automation Conference (DAC '25). doi:10.1109/DAC63849.2025.11132875.
- **Youwei Xiao**, Zizhang Luo, and Yun Liang. 2025. cmt2: Rule-Based Hardware Description in Rust with Temporal Semantics. In 5th Workshop on Languages, Tools, and Techniques for Accelerator Design (LATTE '25). paper.
- Fan Cui, **Youwei Xiao**, Kexing Zhou, and Yun Liang. 2025. An Empirical Comparison of LLM-based Hardware Design and High-level Synthesis. In Proceedings of the 2025 ACM/SIGDA International Symposium on Field Programmable Gate Arrays (FPGA '25). doi:10.1145/3706628.3708861.
- Fan Cui, Chenyang Yin, Kexing Zhou, **Youwei Xiao**, Guangyu Sun, Qiang Xu, Qipeng Guo, Yun Liang, Xingcheng Zhang, Demin Song, and Dahua Lin. 2024. OriGen: Enhancing RTL Code Generation with Code-to-Code Augmentation and Self-Reflection. In Proceedings of the 43rd IEEE/ACM International Conference on Computer-Aided Design (ICCAD '24). doi:10.1145/3676536.3676830.
- **Youwei Xiao**, Zizhang Luo, Kexing Zhou, and Yun Liang. 2024. Cement: Streamlining FPGA Hardware Design with Cycle-Deterministic eHDL and Synthesis. In Proceedings of the 2024 ACM/SIGDA International Symposium on Field Programmable Gate Arrays (FPGA '24). doi:10.1145/3626202.3637561.
- Ruifan Xu, **Youwei Xiao**, Jin Luo, and Yun Liang. 2022. HECTOR: A Multi-Level Intermediate Representation for Hardware Synthesis Methodologies. In Proceedings of the 41st IEEE/ACM International Conference on Computer-Aided Design (ICCAD '22). doi:10.1145/3508352.3549370.

Preprints

- Chenyun Yin*, **Youwei Xiao***, Yuze Luo, Yuyang Zou, and Yun Liang. LLM-Guided Strategy Synthesis for Scalable Equality Saturation. arXiv:2604.17364. (*Equal contribution)
- Zizhang Luo, Yuhao Luo, **Youwei Xiao**, Yansong Xu, Runlin Guo, and Yun Liang. Hive: A Multi-Agent Infrastructure for Algorithm- and Task-Level Scaling. arXiv:2604.17353.
- Yuyang Zou, **Youwei Xiao**, Yansong Xu, Chenyun Yin, Yuhao Luo, Yitian Sun, Ruifan Xu, Renze Chen, and Yun Liang. Aquas: Enhancing Domain Specialization through Holistic Hardware-Software Co-Optimization based on MLIR. arXiv:2511.22267.
- **Youwei Xiao**, Yuyang Zou, and Yun Liang. SkyEgg: Joint Implementation Selection and Scheduling for Hardware Synthesis using E-graphs. arXiv:2511.15323.
- **Youwei Xiao**, Zizhang Luo, Weijie Peng, Yuyang Zou, and Yun Liang. Cement2: Temporal Hardware Transactions for High-Level and Efficient FPGA Programming. arXiv:2511.15073.

Honors and Awards

学术之芯 — 2026
Awarded to 8 students for academic excellence — School of IC, Peking University

博士研究生校长奖学金 — 2026-2027
Awarded to 10 students in School of IC, Peking University — Peking University

ASPLOS 2026 Best Paper Award March 2026
Finding Reusable Instructions via E-Graph Anti-Unification 5/1048 submissions

Second Place in EDAthon 2020 August 2020
Programming Competition on Electronic Design Automation IEEE CEDA Hong Kong Chapter

Shenzhen Stock Exchange Scholarship 2019-2020
Awarded to 17 students from EECS, Peking University

Merit Student 2018-2019 & 2019-2020
Awarded to top 10% students for comprehensive excellence

Yang Fuqing & Wang Yangyuan Academician Scholarship 2018-2019
Awarded to only 5 students at EECS, Peking University, for academic excellence

Second Prize in NOI 2017 July 2017
National Olympiad in Informatics China Computer Federation

Second Prize in APIO 2017 May 2017
Asia Pacific Informatics Olympiad (China District) China Computer Federation

Tutorials

APS: An MLIR-Based Hardware-Software Co-design Framework January 2026
ASP-DAC 2026 Hong Kong, China

Agile Hardware Specialization: A toolbox for Agile Chip Front-end Design May 2025
ISEDA 2025 Hong Kong, China

Agile Hardware Specialization (AHS) March 2025
ASPLOS 2025 Rotterdam, Netherlands

Agile Hardware Specialization: A toolbox for Agile Chip Front-end Design March 2025
DATE 2025 Lyon, France

AHS: An EDA toolbox for Agile Chip Front-end Design January 2025
ASP-DAC 2025 Tokyo, Japan